

Web and Mobile System for Scientific Journal Submission Management

Multidisciplinary Project – 2nd Year

Academic Year 2026–2027

Project Details

Project Type:	2nd Year Multidisciplinary Project
Academic Year:	2026–2027
Number of Students:	2–4 students
Institution:	École Supérieure en Informatique – Sidi Bel Abbès

1. Project Context

Scientific journals need organized procedures for receiving, tracking, reviewing, and managing research manuscripts. In many cases, these activities involve multiple participants, including authors, editors, reviewers, and journal administrators.

This project aims to develop a web and mobile system dedicated to the management of scientific journal submissions. The system will provide a centralized environment through which authors can submit manuscripts and follow their submissions, while editors can manage manuscripts, assign reviewers, and monitor the progress of the review process.

The project will provide students with practical experience in web development, mobile application development, databases, user-interface design, software engineering, and collaborative development.

2. Project Objective

The main objective is to design and implement a web and mobile system that supports the main stages of the scientific journal submission process.

The system should provide a simple, organized, and user-friendly environment for managing manuscripts and their associated information.

3. Specific Objectives

The project objectives include:

1. Analyze the process of scientific journal manuscript submission.
2. Identify the different categories of users and their responsibilities.

3. Design a suitable database for managing users, manuscripts, reviews, and submission status.
4. Develop a web application for journal submission management.
5. Develop a mobile application providing selected system functionalities.
6. Implement user authentication and role management.
7. Provide manuscript submission and document-management functionality.
8. Allow authors to follow the status of their submissions.
9. Provide editors with tools for managing submitted manuscripts.
10. Provide mechanisms for assigning reviewers.
11. Implement a basic peer-review process.
12. Provide notifications concerning important submission events.
13. Develop basic dashboards for the different user roles.
14. Test the system and document its implementation.

4. Target Users

The system will support several categories of users:

- **Author:** submits manuscripts and follows their status.
- **Editor:** manages submissions and the editorial process.
- **Reviewer:** receives assigned manuscripts and submits review reports.
- **Journal Administrator:** manages journal information, users, and system configuration.

5. Main Functionalities

5.1. User Management

The system should provide:

- User registration and authentication.
- Login and logout.
- Password management.
- User profiles.
- Role-based access.

5.2. Author Module

Authors should be able to:

- Create a new submission.
- Enter manuscript metadata.
- Upload manuscript files.
- Add co-authors.
- Modify a submission before editorial processing.
- Follow the status of submitted manuscripts.
- Receive notifications.
- Access previous submissions.

5.3. Editor Module

Editors should be able to:

- View submitted manuscripts.
- Check submission information.
- Assign reviewers.
- Monitor review progress.
- Record editorial decisions.
- Communicate decisions to authors.
- Manage manuscript status.

5.4. Reviewer Module

Reviewers should be able to:

- View assigned manuscripts.
- Accept or decline a review request.
- Access the manuscript assigned for review.
- Submit a review report.
- Provide a recommendation.
- Follow completed reviews.

5.5. Administrator Module

The administrator should be able to:

- Manage users.
- Manage user roles.
- Configure journal information.
- Monitor system activity.
- Manage basic system settings.

6. Submission Process

The system should implement a simplified process:

Submission → Initial Check → Reviewer Assignment → Peer Review →
Editorial Decision → Final Status

Possible submission statuses include:

- Submitted.
- Under Initial Check.
- Under Review.
- Revision Needed.
- Accepted.
- Rejected.

7. Web and Mobile Applications

The project will consist of two complementary interfaces.

7.1. Web Application

The web application will provide complete access to the system and will be primarily intended for authors, editors, reviewers, and administrators.

7.2. Mobile Application

The mobile application will provide selected functionalities such as:

- Authentication.
- Submission-status monitoring.
- Notifications.
- Review requests.
- Review-status monitoring.
- Access to important submission information.

The mobile application does not necessarily need to reproduce every functionality of the web application.

8. System Architecture

Students should design a modular architecture containing:

- User interface layer.
- Application or business-logic layer.
- Database layer.
- Authentication and authorization components.
- File-management component.
- Notification component.

A REST API may be used to connect the web and mobile applications to a common backend.

9. Database Design

The database should contain appropriate entities such as:

- Users, Roles, Authors, Manuscripts, Authors associated with manuscripts, Reviewers, Review assignments, Reviews, Editorial decisions, Notifications, Submission status history.

Students should produce an entity–relationship model before implementation.

10. Proposed Technologies

Students may use technologies such as:

- HTML, CSS, and JavaScript.
- React, Angular, or Vue for the web interface.
- Flutter or React Native for the mobile application.
- Node.js, Django, Spring Boot, Laravel, or another suitable backend framework.
- MySQL, PostgreSQL, or another relational database system.
- REST API for communication between applications and the backend.
- Git and GitHub/GitLab for version control.

The final technology selection should be agreed upon by the project team.

11. Expected Deliverables

The team should provide:

1. Specifications document.
2. System architecture.
3. UML diagrams.
4. Database conceptual and logical models.
5. Functional web application.
6. Functional mobile application.
7. Backend and REST API.
8. Database implementation.
9. Authentication and role management.
10. Submission-management process.
11. Basic notification system.
12. Test plan and test results.
13. Source code.
14. Technical documentation.
15. User guide.
16. Final project report.
17. Final demonstration.

12. Skills and Competencies

Through this project, students will develop practical skills in:

- System Specification and Design.
- Database design.
- Web application development.
- Mobile application development.
- Backend development.
- API design.
- User-interface design.
- Authentication and authorization.
- Software testing.
- Version control and collaborative development.
- Technical documentation.

13. Possible Future Extensions

The system may later be extended with:

- Automated manuscript format checking, Plagiarism detection integration, ORCID integration, Email integration, Advanced reviewer recommendation, Editorial statistics, Researcher profiles, Automated report generation.

14. Keywords

Scientific Journal, Manuscript Submission, Peer Review, Web Application, Mobile Application, Database, REST API, Software Engineering, Scientific Publishing.

Contact

Students interested in this project are invited to contact:

Dr. Mohammed Bekkouche
École Supérieure en Informatique – Sidi Bel Abbès (ESI-SBA)
LabRI-SBA Laboratory
Email: m.bekkouche@esi-sba.dz